

Mandibular organ: it's relation to body weight, sex, molt and reproduction in the crab, *Oziotelphusa senex senex* Fabricius (1791)

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Abstract

The dynamics of the mandibular organ (MO) in the edible fresh water field crab *Oziotelphusa senex senex* was studied as a function of sex, weight, reproduction, molting and eyestalk ablation (ESX). The weight of the mandibular organ exhibited a positive correlation with the body weight. In general, male crabs possess larger mandibular organs when compared to female crabs. The weight of the mandibular organ increased with ovarian index, molting and as the crab progressed towards reproduction. Bilateral eyestalk ablation resulted in a significant increase in mandibular organ weight. The methyl farnesoate (MF) content of the mandibular organ increased with increase in organ weight. The results presented strongly support a potential role of the mandibular organ in regulating both molting and reproduction.

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1. Introduction

The mandibular organs (MOs) in crustaceans are paired, ductless, highly vascularized, ectodermally derived glands, which are located at the base of the mandibular tendons. These organs were first described by Le Roux (1968) and are often confused with the Y-

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organs, because of the close anatomical proximity and histological similarities. Both of the organs show ultrastructural and cyclical changes correlated with the molt stage of the animal (Auto et al., 1974; Byard, 1975). The MOs can now be conveniently distinguished from the Y-organs biochemically, since the former synthesize and secrete methyl farnesoate (MF) and the latter ecdysteroids. The physiological function for MF in crustaceans is less well known, but evidence is accumulating to support a role for MF in growth (Yudin et al., 1980) and reproduction (Laufer et al., 1993; Jo et al., 1999). MF synthesis is regulated by an eyestalk neuropeptide (Borst et al., 1988; Landau et al., 1989). A mandibular organ inhibiting hormone (MOIH), secreted by the sinus gland in the eyestalks, has been identified and sequenced in some crustacean species (Liu et al., 1997a,b; Webster, 1998).

In order to grow and to increase in size, crustaceans must periodically replace their protective exoskeletons with larger ones by a process called molting, which is under hormonal control. 20-Hydroxyecdysteroid is the molting hormone in crustaceans. Ecdysone, the precursor for 20-hydroxyecdysteroid, is synthesized and secreted from the Y-organs (Gabe, 1953). Ecdysteroid is regulated at the level of synthesis and secretion by a neuropeptide hormone namely molt inhibiting hormone present in the eyestalk X-organ-sinus gland complex. Eyestalk ablation causes an increase in ecdysteroid secretion, suggesting that there are inhibitory factors in the eyestalk that suppress ecdysteroid secretion.

In the fresh water crab *Oziotelphusa senex senex* Fabricius, molting and ovarian maturation do not occur simultaneously, with the full cycle of ovarian development occurring during intermolt stage of the molt cycle (Reddy et al., 1985). Because these processes are highly energy demanding, the animal may choose either to grow or to reproduce depending on the complex external and internal factors. Since MF is known to regulate both molting and reproduction in crustaceans, in the present study, we investigated the dynamics of the mandibular organ in the crab *O. senex senex* under different physiological conditions. The MF content in different weight MOs was also measured.

2. Materials and methods

Adult Indian rice field crabs, *O. senex senex* Fabricius (1791), were collected from paddy fields in and around Tirupati (Andhra Pradesh, India) and maintained in the laboratory at 25 ± 1 °C in fresh water tubs. They were acclimatized to laboratory conditions (12:12 L/D) for 1 week before being used in experiments.

The crabs were immobilized by chilling on ice for 10 min. The body weights of the crabs were determined. The MOs from both sides and reproductive organs were isolated, immediately placed in ice cold crustacean physiological saline (Van Harreveld, 1936) to scrap off adhesive tissue. The organs were removed from the saline, lightly blotted with paper towels and weighed wet on an electronic balance. The reproductive stages were identified according to Reddy and Ramamurthi (1998), and the molt stages were observed using setal development in the mastigobranch of third maxillipede as described for this crab (Reddy, 1991).

The gonadal indices were determined using the standard formula:

$$\text{gonad index} = \frac{\text{Wet weight of the gonad}}{\text{Wet weight of the crab}} \times 100$$

Eyestalks were removed from the crabs by cutting them off at the base without prior ligation but with cautery of the wound after operation. The eyestalk-ablated crabs were sacrificed after 5 days and MOs were isolated. The MO index was determined using the following formula:

$$\text{MO index} = (\text{MO weight/body weight}) \times 100$$

MF content in the MOs was estimated essentially as described by Borst and Tsukimura (1991). In brief, the MOs were homogenized in 199 medium with Earle's salts and extracted three times in hexane and lyophilized. The concentration of MF was determined using HPLC and authenticated by using MF standard (obtained from Prof. R. Ramamurthi).

The significance of differences between the means of different size glands was determined by using one-way ANOVA followed by Student–Newman–Keuls test.

3. Results

In Figs. 1 and 2, data on the weight of the MO in different weight crabs are reported. As shown, the weight of the MO increased with increasing body weight in both male and female crabs. From the regression model, the MO mass increased on an average by 0.485 mg/g body weight in males and 0.49 mg/g body weight in females. The results also demonstrate that the MO weight is under the negative regulation of eyestalk contents

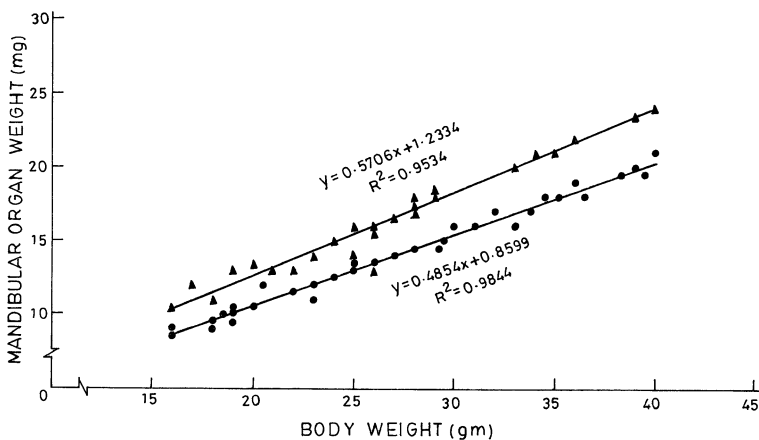


Fig. 1. Changes in mandibular organ weight in different sized male crabs. Closed circles represent control males and closed triangles represent eyestalk-ablated males.

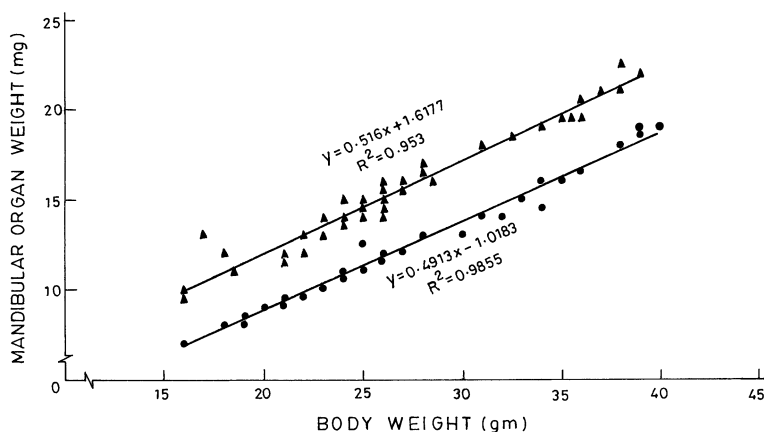


Fig. 2. Changes in mandibular organ weight in different sized female crabs. Closed circles represent control females and closed triangles represent eyestalk-ablated females.

(Figs. 1 and 2). Bilateral eyestalk ablation (ESX) significantly increased MO weight in both male and female crabs. From the regression model, the weight of the MO increased on an average by 0.571 mg/g body weight in ESX males and 0.51 mg/g body weight in ESX females in 5 days.

The MO weight in male crabs is significantly ($p < 0.001$) greater than the female crabs (Figs. 1 and 2). The results presented in Table 1 show that the weight of the MO increased with advancement of reproductive stage of female crab. The MO weight increased significantly as the female crab approached an advanced reproductive stage (Vitelloginesis: Stage III) and also exhibited positive correlation ($r = 0.2908$) with ovarian index (Table 1). On the contrary, the MO weight did not show any correlation with testicular index (authors' unpublished data).

Table 1

Changes in ovarian index and mandibular organ (MO) weight and MO index during different reproductive stages of the crab *O. senex senex*

Reproductive stage	Color of the ovary	Weight of the crab	Ovarian index	MO weight (mg)	MO index
Immature ($N=47$)	White	20.28 ± 3.313	0.258 ± 0.089	8.43 ± 1.30	0.416 ± 0.065
Vitellogenic stage I ($N=12$)	Pale yellow	22.21 ± 2.444 (9.52)	0.388 ± 0.080 (50.39)	9.38 ± 0.53 (11.27)	0.422 ± 0.058 (1.44)
Vitellogenic stage II ($N=9$)	Orange	22.61 ± 3.060 (11.49)	0.576 ± 0.159 (123.26)	10.56 ± 0.88 (25.27)	0.467 ± 0.076 (12.25)
Vitellogenic stage III ($N=5$)	Dark brown	24.10 ± 1.884 (18.84)	1.631 ± 0.22 (532.02)	13.70 ± 0.45 (62.51)	0.568 ± 0.044 (36.53)
<i>F</i> ratio		1.616	288.64	38.167	9.394
<i>p</i> value		0.194	<0.0001	<0.0001	<0.0001

Values are mean $\pm$ S.D.

Values in parentheses are % change from immature crabs.

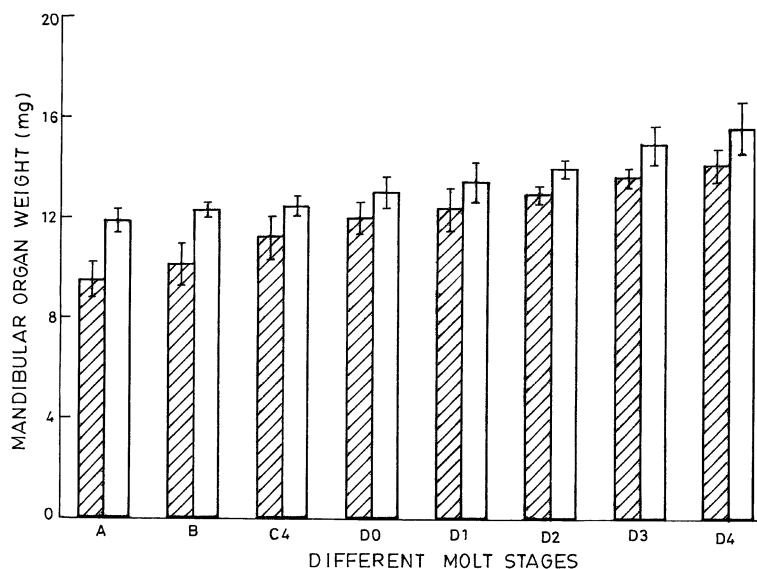


Fig. 3. Mandibular organ weight in different molt stages of the crab, *O. senex senex*. Open bars represent mandibular organs in male crabs (body weight ranged from 18.3 to 21.7 g) and striped bars represent mandibular organ weights in female (body weight ranged from 19.0 to 21.5 g) crabs. Vertical bars represent standard deviation. For all the groups $n=15$.

In Fig. 3, data are reported on the weight of the MO in crabs at different molt stages. For this experiment, crabs in the weight range of 18–22 g were used (Fig. 3). As shown, the mean MO weight increased progressively as the animal approached actual ecdysis, whereas the weight of MOs between each stage showed no significant change. Among all the molt stages, postmolt crabs contain small size mandibular organs.

The MF content was reported in Table 2 for different weight mandibular organs collected from intermolt, previtellogenic stage crabs. It is clear from the data that the MF content increased with increase in size of mandibular organ. (F ratio=6.307; p value is 0.0194). The rate of synthesis of MF (ng MF/g MO) is more in small size glands when compared to large size mandibular organs.

Table 2

Methyl farnesoate content in different size mandibular organs of the crab, *O. senex senex*

Weight of the mandibular organ (mg)	MF content (ng)/MO	Rate of MF synthesis (ng MF/mg MO)
8 ± 0.91	220 ± 32	27.5
12 ± 0.82	280 ± 36	23.3
16 ± 0.91	320 ± 50	20
F ratio	6.307	4.376
p value	0.0194	0.0186

Values are mean $\pm$ S.D. ($n=4$).

Only intermolt, previtellogenic female crabs were used in the present study.

Methyl farnesoate content was estimated using single mandibular organs.

4. Discussion

Since the discovery of MF in crustaceans (Laufer et al., 1987a), the physiological role of this compound has been subject to debate (Laufer et al., 1993; Homola and Chang, 1997; Reddy and Ramamurthi, 1998). Methyl farnesoate is known to be involved in the regulation of molting (Yudin et al., 1980; Chang et al., 1993; Homola and Chang, 1997; Tamone and Chang, 1993; Wilder et al., 1995; Tamone et al., 1997), some aspects of reproduction (Laufer et al., 1987b, 1993; Sagi et al., 1993; Laufer, 1992; Ahl et al., 1996; Ahl and Laufer, 1996; Reddy and Ramamurthi, 1998; Jo et al., 1999; Rodriguez et al., 2002), morphogenesis (Laufer et al., 1997; Abdu et al., 1998; Rotllant et al., 2000) and general protein synthesis (Paulson and Skinner, 1988; Soroka et al., 1993).

The amount of MF detected in the MOs of *O. senex senex* was in the range of 220 ng/MO which was very similar to those reported for the edible crab, *Cancer pagurus* (Wainwright et al., 1996a), but higher than the concentrations reported for the American lobster *Homarus americanus* (3.2–196.8 ng/MO) (Borst et al., 1994). As the weight of the gland increased, the amount of MF in the MO increased. Bilateral eyestalk ablation increased MO weight. Hinsch (1977) and Le Roux (1980, 1983) also observed hypertrophy of the MO in response to eyestalk ablation in *Libinia emarginata*, *Palaemonetes varians* and *Pisidia logicornis*. Eyestalk removal significantly increased the levels of MF in the hemolymph of spider crab *L. emarginata* (Laufer et al., 1987b), American lobster *H. americanus* (Tsukimura and Borst, 1992) and *Orconectes virilis* (Tsukimura et al., 1989). Eyestalk removal also caused a 20–100 fold increase in methyl transferase activity, which mediates the final step of MF synthesis in the MO of crayfish *Procambarus clarkii* (Landau et al., 1989). Recently, a mandibular organ inhibiting hormone (MOIH) has been isolated from the sinus gland (Liu and Laufer, 1996), sequenced (Liu et al., 1997a) and cloned (Liu et al., 1997b; Wainwright et al., 1996b). MOIH is a member of the crustacean hyperglycemic hormone family and it shares 50–60% sequence similarity with the members of this group, particularly with gonad inhibiting hormone and molt inhibiting hormone (Wainwright et al., 1996b).

MO weight increased significantly during reproduction in the crab *O. senex senex*. Hinsch (1980) observed that the increase in gonad size is stimulated by MO implantation in *L. emarginata*. Using *P. clarkii*, Laufer et al. (1998) demonstrated that MF promotes ovarian maturation. Stimulation of ovarian maturation by exogenous MF was also observed in the crab *O. senex senex* (Reddy and Ramamurthi, 1998). The MF synthesis by MOs is lowest in juvenile and previtellogenic female, highest during vitellogenesis (Laufer et al., 1986, 1987a). However, Wilder et al. (1995) observed no correlation between MF titre in the hemolymph and reproductive cycle; whereas concentration MF increased as the animal approached to ecdysis. It has been demonstrated that mandibular gland homogenate stimulates molting in *Callinectes sapidus* (Yudin et al., 1980). Co-incubation of MO with Y-organs stimulated ecdysteroid secretion in *Cancer magister* in vitro (Tamone and Chang, 1993) also indicates that the MO was responsible for the stimulation of a precocious molt, via an in vivo stimulation of secretion of ecdysteroids from Y-organs. In the crab *O. senex senex* also, the weight of the MO increased as the crab approached ecdysis. Administration of MF also caused molt acceleration in *Cherax quadricarinatus* (Abdu et al., 2001).

The MOs of male crabs were larger than those of female crabs. Similar results were observed for the spider crab *L. emarginata* (Laufer et al., 1987b), crayfish *P. clarkii* (Landau et al., 1989) and fresh water prawn *Macrobrachium rosenbergii* (Sagi et al., 1991). The physiological significance of larger MOs in male crabs is not known at present, but it is possible that MOs may be involved in the regulation of male-specific activities in these animals.

The X-organ sinus gland complex in the eyestalk synthesizes two groups of peptides, the crustacean hyperglycemic hormone (CHH) neuropeptide family and the chromatotrophins (Van Herp and Soye, 1997). Stimulation of the MO of crayfish *P. clarkii* to synthesize MF by chromatographic red pigment-concentrating hormone was reported earlier (Landau et al., 1989). The neuropeptides belonging to the CHH family (hyperglycemic hormone, gonad inhibiting hormone, molt inhibiting hormone and mandibular organ inhibiting hormones) are known to be involved in the synthesis and secretion of ecdysteroids and MF, regulating growth and reproduction (Van Herp, 1998). CHH receptors have been found in the Y-organ membrane preparations suggesting a role for hyperglycemic hormone in regulation of ecdysteroid synthesis (Webster and Keller, 1986). A peptide characterized as a functional molt inhibiting hormone (MIH) in American lobster *H. americanus* is also functionally hyperglycemic (Chang et al., 1990). Webster (1998) discovered that MIH and MOIH are completely co-localized in the X-organ though functionally distinct. The overlapping bio-activities and the role of these neuropeptides are to be determined in the production of ecdysteroids and MF.

Eyestalk ablation removes the source of inhibition of the MO, resulting in increased MF levels in the hemolymph of *L. emarginata* (Laufer et al., 1986, 1987b) and of *H. americanus* and *O. virilis* (Tsukimura et al., 1989). Eyestalk removal causes hypertrophy and changes in the ultrastructure of the MO (Byard et al., 1975; Bazin, 1976; Hinsch, 1977; Le Roux, 1983), further showing that the MO is stimulated by eyestalk ablation. Tamone and Chang (1993) reported that ecdysteroid secretion was stimulated by MF in *C. magister*. Previous studies have demonstrated MF induced delay (Borst et al., 1987; Ahl and Brown, 1990) or disruption (Abdu et al., 1998) in larval development. Whether MF is directly involved in the regulation of molting or acts by stimulating the ecdysteroid production by the Y-organs in the crab *O. senex senex* remains to be studied.

Taken together with what now appears to be a multi-functional role for MF, our results suggest that like many other crustacean hormones, MF has at least dual functions—involved in the regulation of molting and reproduction. Further studies are needed to verify the other physiological roles of MF in crustaceans. Since it is known that relatively large MOs are present during molting and reproduction (Hinsch, 1977; Yudin et al., 1980), it is possible that MOs also control related processes such as regulation of salt and water balance, mineral metabolism, vitellogenin synthesis, etc. Further experiments are in progress to establish the other physiological roles of the mandibular organ.

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